Reg. No. | | | | | | | | | | | [12EC53/13EC53]

FIFTH SEMESTER B.E/B.TECH EXAMINATION DECEMBER-2015

DIGITAL SIGNAL PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Derive the relationship between **08**

i. DFT and Z Transform

ii. DFT and DFS

b. Compute the following **06**

i. N point DFT of $x(n)=a^n$ for $o \le n \le N-1$

ii. Determine IDFT of X(k)={3,2+j,1,2-j}

2. a. g(n) and h(n) are the two sequence of length 6. They have 6 point DFT(s) G(k) and H(k) respectively. The sequence g(n) is given as g(n)={4.1, 3.5,1.2,5,2,3.3}. The DFTS G(k) and H(k) are related by the circular frequency shift are H(k)= G((k-3))6. Determine h(n) without computing DFT of IDFT. **08**

b. The five samples of the 8 point DFT **06**

X(0)=0.25, X(1)=0.125-j0.3018

X(6)=X(4)=0, X(5)= 0.125-j0.0518

Determine the remaining samples if the sequence x(n) is real valued sequence.

3. a. Why FFT is needed. What is the speed improvement factor in calculating 64 point DFT of a sequence using Direct Computation and FFT algorithm. **06**

b. Find the output y(n) of a filter whose impulse response is h(n)= {1,1,1} and the input to the filter is x(n)= {3, -1, 0, 1, 3, 2, 0, 1, 2, 1} using overlap save method. **08**

4. a. Use 8 point DIF-FFT radix 2 algorithm to find the DFT of the sequence X(n) {1, -1, -1, -1, 1, 1, 1, -1} **08**

b. Describe Chirp Z transform algorithm to compute Discrete transform of a given sequence. **06**

5. a. Design a Chebyshev analog lowpass filter that has -3db cutoff frequency of 100rad/ sec and a stop band attenuation of 25 db or greater for all radian frequencies past 250 rad/sec. **09**

b. Write a short note on analog to analog frequency transformation. **05**

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6. a. Design a digital Chebyshev type I filter that satisfies the following constraints. **14**

$0.8 \leq |H(\omega)| \leq 1,\ 0 \leq \omega \leq 0.2\pi$

$|H(\omega)| \leq 0.2,\ 0.6\pi \leq \omega \leq \pi$

Use impulse invariant transformation.

7. a. A low pass filter is to be designed with the following desired frequency response **08**

$$H_d(e^{ow}) = H_{d(w)} = \begin{cases} e^{-j2w} & |w| < \pi/4 \\ o & \pi/4 < |w| < \pi \end{cases}$$

Determine the filter coefficient $h_d(n)$ and h(n), if w(n) rectangular window defined as follows.

$W_R(n) = \begin{cases} 1,\ 0 \leq n \leq 4 \\ 0,\ otherwise \end{cases}$ Also find frequency response H(w).

b. A low pass filter has the desired frequency response **06**

$$H_d(w) = H_d(e_{sw}) = \begin{cases} e^{-j3w}, 0 < w < \pi/2 \\ 0, \quad \pi/2 < w < \pi \end{cases}.$$

Determine h(n) based on frequency sampling techniques. Take N=7

8. a. Obtain cascade realization for a system having following system function **07**

$$H(z) = \frac{(z-1)(z-2(z+1)z}{\left(z-\frac{1}{2}-j\frac{1}{2}\right)\left(z-\frac{1}{2}+j\frac{1}{2}\right)\left(z-j\frac{1}{4}\right)\left(z+j\frac{1}{4}\right)}.$$

b. Realize the linear- phase FIR filter having the following impulse response. **07**

$$h(n) = \delta(n) + \frac{1}{4}\delta(n-1) - \frac{1}{8}\delta(n-2) + \frac{1}{4}\delta(n-3) + \delta(n-4).$$